

International Transfers or Differentiated Carbon Prices?

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Abstract

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1 Introduction

The Paris Agreement commits its signatories to limiting global warming to 1.5–2°C above pre-industrial levels. Meeting that target requires cutting global CO₂ emissions by roughly 28 GtCO₂ per year relative to current policy trajectories, yet the 2025 round of Nationally Determined Contributions (NDCs) leaves expected 2030 emissions largely unchanged (Olhoff et al. 2025). Worse still, global CO₂ emissions actually grew by 2.3 per cent in 2024, nearly four times the 0.6 per cent average growth rate of the 2010s. This shortfall reflects a distributional disagreement over who should bear the cost of decarbonisation, a disagreement that international climate negotiations have never cleanly resolved.

At the heart of this disagreement lies a choice between two families of instruments. One prescribes a single, uniform carbon price applied everywhere, with any resulting inequities corrected through separate financial transfers. The other lets, or requires, each country to set its own carbon price according to its income and development needs, in lieu of transfers that have so far not materialized. The two camps rarely engage with each other’s central premise: advocates of differentiated prices treat the infeasibility of large international transfers as a fact of political life, while advocates of uniform pricing treat the efficiency of a single global price as decisive and leave transfers to be sorted out separately. This paper argues that the disagreement is, to a first approximation, a false one. For any system of nationally differentiated carbon prices, there exists a uniform global price combined with a specific allocation of emission rights, that is, a particular pattern of transfers, that leaves every country exactly as well off. Because abatement costs are convex, that uniform-price allocation is moreover weakly more efficient for every country simultaneously. The debate over prices versus rights is therefore less a debate about fundamentally different instruments than about how the gains from a single instrument should be distributed.

We formalise this equivalence in a static, single-period model, then ask whether it extends to the dynamic, multi-period setting a real climate policy requires. The equivalence holds cleanly in the static case. It breaks down once time is introduced, because the welfare value of an emission right then depends not only on its quantity but on the full trajectory of the future carbon price, a trajectory that cannot be recovered analytically without knowledge of how capital markets, abatement technologies, and emissions elasticities evolve. We characterise two special cases that restore a partial equivalence: the case of homothetic autarky prices with fixed budget shares, and the Hotelling case, in which efficient capital markets and perfect foresight keep discounted carbon prices constant over time. In the Hotelling case, the dynamic formula collapses to one that is structurally identical to the static result, with the intertemporal carbon budget playing the role of the single-period emission right.

We then test the equivalence numerically using the NICE integrated assessment model (Young-Brun et al. 2025), which disaggregates regional populations into income quintiles and therefore allows precise measurement of distributional effects. For each of eight representative countries or regions, we compute the non-losing rights allocation ρ_1 , the

minimum rights multiplier at which the country weakly prefers the uniform regime, and compare it to the static formula’s prediction. The static formula explains the cross-country variation in ρ_1 well as a first approximation: the non-losing allocation is roughly predicted by each country’s per-capita emissions relative to the world average, consistent with the equivalence result. The three largest deviations from this prediction reveal distinct mechanisms that the static framework abstracts from: general-equilibrium feedback through the rest of the world for China, prior abatement-cost exhaustion for the European Union (27 states), and a rapidly growing emission baseline for India. In all eight cases, there exists a rights allocation above which the uniform-price regime weakly dominates autarky on efficiency grounds, and that threshold is closely predicted by the static formula.

The remainder of the paper is organised as follows. Section 2 lays out both sides of the disagreement, together with the existing efficiency- and equity-based arguments for price differentiation. Section 3 derives the static distributional equivalence and the Diamond–Mirrlees efficiency result, then characterises the conditions under which a partial equivalence extends to the dynamic case. Section 4 describes the simulation design and presents the results, assessing where the static prediction holds and where it fails. Section 5 applies the framework to two recent differentiated-pricing proposals, computing the rights allocation each would require to be welfare-equivalent to uniform pricing. Section ?? concludes.

2 Rationales for differentiated carbon prices

Lower-income economies, whose per-capita emissions are below the global average, generally support the case for differentiation on two grounds: a high uniform carbon price would impose disproportionate adjustment costs on populations still building out basic infrastructure and industrial capacity, and historical responsibility for cumulative emissions lies overwhelmingly with wealthier economies. On these grounds, they claim the right to set lower domestic prices, or to delay ambitious pricing until adequate compensation is on the table. High-income economies and most economists counter that a uniform price minimises the global cost of any given amount of abatement, and that equity is better handled by paying for redistribution directly rather than through the price itself.

Setting aside the equity case for the moment, three distinct arguments from the theory of the second-best give differentiation a purely efficiency-based justification, independent of any distributional concern. First, [Anthoff and Emmerling \(2022\)](#) show that countries with different growth rates face different discount rates, so that even a purely welfare-maximising social planner, with no political constraints at all, would set carbon prices that differ across countries. Second, [Babiker \(2005\)](#) demonstrate theoretically, and [Boeters \(2014\)](#) confirm numerically, that pre-existing distortions, such as market power in traded goods and sector-specific taxes, can make a differentiated price pattern more efficient than a uniform one; [Boeters \(2014\)](#) estimate the potential gain from optimal differentiation at the equivalent of a 27% emission reduction for free. Third, and closest to the concern of

this paper, is the claim that large international financial transfers are politically infeasible, which makes differentiated prices a necessary second-best substitute for redistribution (Bauer et al. 2020; Young-Brun et al. 2025).

It is this third, transfer-infeasibility argument that our equivalence result speaks to directly. The first two arguments rest on distortions that a uniform-price regime cannot correct by construction, whatever allocation of rights accompanies it. Our result has nothing to say about them, and Section 3’s efficiency result is correspondingly a first-best statement that holds only in their absence. The third argument, by contrast, treats transfers and prices as separate policy levers, one infeasible and the other not. But an allocation of emission rights under a cap-and-trade is itself a transfer, denominated in tonnes rather than currency, and cleared through the same market that sets the uniform price. If any given differentiated-price outcome can be reproduced by a suitable rights allocation, as we show below, then the political constraints that make direct cash transfers hard to negotiate would have to apply to emission rights as well for the infeasibility argument to retain its force; it cannot be invoked as a reason to prefer differentiated prices over trading per se.

3 Correspondence between transfers in a cap-and-trade and differentiated carbon prices

This section establishes the core theoretical claims of the paper. We begin with a static, single-period model in which the equivalence between differentiated prices and uniform prices with transfers takes a closed, analytically tractable form. We then extend the analysis to a dynamic setting and show that the correspondence breaks down, because the welfare value of an emission right depends on the price trajectory rather than on quantities alone. We characterise two special cases that restore a partial equivalence, homothetic prices with a fixed budget share, and the Hotelling case, and explain why the Hotelling case serves as the natural empirical benchmark for the simulations that follow.

3.1 The static case

Setting. Consider a world of many countries indexed by i , a single period, and no uncertainty. Every country has an incentive to reduce its emissions, but decarbonisation is costly. When country i faces a carbon price p_i , firms substitute towards low-carbon capital and households shift to less preferred, less carbon-intensive consumption. We call these abatement costs and denote them a_i . Allowing for international transfers t_i (positive if received, negative if paid) and abstracting from exogenous investment, country i ’s per-capita consumption is:

$$c_i = y_i + t_i - a_i, \tag{1}$$

where y_i is potential output per capita.

Welfare of country i depends on its consumption c_i and on global emissions $E = \sum_j e_j n_j$, where e_j is country j 's per-capita emissions and n_j its population. We fix global emissions at their optimal level E^* : the question we ask is not “what is the optimal target?” but “given an agreed global target, which policy regime distributes its costs and benefits most fairly and efficiently?” The temperature outcome is identical under both regimes, only distribution and efficiency differ.

Let p^* be the uniform carbon price that attains E^* , defined implicitly by $E^* = \sum_j e_j(p^*) n_j$. We use the shorthands $e_i := e_i(p_i)$, $a_i := a_i(e_i)$, $e_i^* := e_i(p^*)$, and $a_i^* := a_i(e_i^*)$. A situation for country i , given fixed global emissions, can be summarised as $s = (t_i, p_i)$, and we write welfare as $u_i(t_i, p_i)$ with a slight abuse of notation.

The equivalent variation from the benchmark. To analyse an arbitrary situation s , it is useful to anchor the analysis to two special cases. The first is the benchmark situation $s^* = (0, p^*)$: no transfers and a uniform carbon price. The second is the autarky situation $s^a = (0, p_i)$: no transfers and a nationally differentiated price. We define $V_i(p_i)$ as the equivalent variation between autarky and the benchmark. It is the lump-sum transfer that, if paid at the benchmark price p^* , would make country i exactly as well off as under autarky:

$$u_i(V_i(p_i), p^*) = u_i(0, p_i). \quad (2)$$

Since consumption under autarky is $c_i^a = y_i - a_i$ and under the benchmark is $c_i^* = y_i - a_i^*$, we have:

$$V_i(p_i) = c_i^a - c_i^* = a_i^* - a_i. \quad (3)$$

This difference in abatement costs can be approximated by a first-order Taylor expansion around the benchmark emissions level e_i^* . At e_i^* , the optimising country sets its marginal abatement cost equal to the price it faces. Under p^* this gives $\frac{da_i}{de}(e_i^*) = -p^*$. Therefore:

$$V_i(p_i) \approx (e_i - e_i^*) p^*. \quad (4)$$

Denoting by $\varepsilon^* = -\frac{de_i/e_i^*}{dp/p^*}$ the price elasticity of emissions evaluated at p^* , we can write the equivalent variation more explicitly as:

$$V_i(p_i) \approx (p^* - p_i) e_i^* \varepsilon^*. \quad (5)$$

The sign is intuitive: if $p_i < p^*$, country i abates less in autarky and therefore has lower abatement costs. Moving it to the benchmark forces higher abatement, so a positive compensation $V_i > 0$ is required to hold welfare constant. The magnitude of the compensation is larger when the price gap is large, when baseline emissions are high, and when emissions are price-elastic. All three make the cost of tightening the price more burdensome.

The equivalent transfer for an arbitrary situation. Let g_i^s denote country i 's welfare gain in situation $s = (t_i, p_i)$ relative to the benchmark, expressed as a money-equivalent transfer at p^* . We call this the equivalent transfer. It decomposes cleanly as:

$$g_i^s = t_i + V_i(p_i). \quad (6)$$

The first term is the actual transfer received, the second is the welfare gain or loss from facing a price that differs from the benchmark. Because g_i^s reduces every situation to a single money-equivalent number at a common price, it is the natural metric for comparing welfare across regimes that differ both in transfers and in prices.¹

Correspondence between differentiated prices and emission rights. We can now state the central result of the static case. Suppose countries establish a cap-and-trade at the uniform price p^* , and each person in country i is allocated emission rights r_i . The net monetary transfer to i under the cap-and-trade is:

$$t_i^* = (r_i - e_i^*) p^*. \quad (7)$$

Country i sells rights if $r_i > e_i^*$ (it was allocated more than it emits at p^*) and buys them otherwise. The equivalent transfer in the cap-and-trade situation is therefore $g_i^* = t_i^*$. The equivalent transfer in autarky is, from equation (5):

$$g_i^a = V_i(p_i) \approx (p^* - p_i) e_i^* \varepsilon^*. \quad (8)$$

Setting $g_i^* = g_i^a$ and solving yields the equivalence relation:

$$\boxed{(r_i - e_i^*) p^* \approx (p^* - p_i) e_i^* \varepsilon^*,} \quad (9)$$

from which one can isolate either variable:

$$r_i \approx \left(1 + \left(1 - \frac{p_i}{p^*}\right) \varepsilon^*\right) e_i^*, \quad (10)$$

$$p_i \approx \left(1 + \frac{e_i^* - r_i}{e_i^* \varepsilon^*}\right) p^*. \quad (11)$$

Equation (10) is the key formula for policy design: given a country's autarky price p_i , it tells us how many emission rights to allocate in a cap-and-trade in order to leave that country's welfare unchanged. A country with a low autarky price ($p_i < p^*$) needs a generous rights allocation ($r_i > e_i^*$) to compensate for the abatement burden it takes on when forced to the higher uniform price; a country with a high autarky price ($p_i > p^*$) can be left with fewer rights than its equilibrium emissions.

¹One can also define an equivalent price p_i^* such that $(0, p_i^*)$ is welfare-equivalent to (t_i, p_i) . Solving $t_i + (p^* - p_i) e_i^* \varepsilon^* \approx (p^* - p_i^*) e_i^* \varepsilon^*$ gives $p_i^* \approx p_i - t_i / (e_i^* \varepsilon^*)$. This equivalent price can be negative when the transfer is large, however, which has no natural economic interpretation as a carbon price. The equivalent transfer g_i^s does not share this limitation, which is why we use it as our primary welfare metric.

This equivalence result shows that the two regimes can, in principle, deliver identical welfare distributions: any differentiated-price arrangement corresponds to a cap-and-trade with a specific rights allocation, and vice versa. This raises a natural follow-up question. If the two regimes can always be made equivalent by choosing the right allocation, is there any reason to prefer one over the other? We now show that the answer is yes: the uniform-price regime strictly dominates differentiated prices on efficiency grounds, for every country simultaneously, whenever abatement cost functions are strictly convex. The argument is a direct application of the Diamond–Mirrlees production efficiency theorem, and its mechanism is straightforward: under differentiated prices, countries at a lower domestic price forgo abatement whose cost is below the market rate. Under a cap-and-trade, those same reductions are sold at the full market price p^* , which is always at least as favourable a deal.

Efficiency of the uniform price. Suppose we give country i rights equal to its autarky emissions e_i , so that the net transfer under the cap-and-trade is $\tilde{t}_i = (e_i - e_i^*)p^*$. Convexity of abatement costs implies:

$$a_i - a_i^* \geq (e_i - e_i^*) \frac{da_i^*}{de} = (e_i - e_i^*)(-p^*), \quad (12)$$

and therefore $\tilde{t}_i = (e_i - e_i^*)p^* \geq a_i^* - a_i = V_i(p_i) = g_i^a$. Country i 's consumption under the cap-and-trade is:

$$\tilde{c}_i = y_i + \tilde{t}_i - a_i^* \geq y_i - a_i = c_i^a. \quad (13)$$

Welfare is therefore weakly higher under the cap-and-trade than in autarky, for every country simultaneously. In autarky, a lower price reduces abatement costs, but at a diminishing rate, because of convexity. Selling spare permits on the cap-and-trade market generates revenue at the full market price p^* , which is always at least as good a deal as facing a lower domestic price. Trading strictly dominates autarky whenever abatement costs are strictly convex and prices differ across countries.

This is a statement of the Diamond–Mirrlees production efficiency theorem [Diamond and Mirrlees \(1971\)](#) applied to international climate policy: in the absence of other distortions, direct transfers are always preferable to differentiated factor prices as a redistributive instrument.

3.2 The dynamic case

Setting. We now add a time index t running over multiple periods. Each variable becomes a trajectory: prices $(p_{it})_t$, transfers $(t_{it})_t$, emissions $(e_{it})_t$. Let β_t be the discount factor between the initial period and period t . Intertemporal quantities are written in upper case: $G_i = \sum_t g_{it}\beta_t$ is the total discounted equivalent gain, and $R_i = \sum_t r_{it}$ is country i 's intertemporal carbon budget.

Welfare equivalence now requires equality of entire discounted streams rather than point-in-time values. The natural dynamic extension of the static formula (9) is:

$$G_i^U = G_i^A \iff \sum_t (r_{it} - e_{it}^*) \beta_t p_t^* \approx \sum_t (p_t^* - p_{it}) e_{it}^* \varepsilon_t^* \beta_t. \quad (14)$$

In the static case, equation (9) admits a clean solution because p^* appears as a common factor on both sides and cancels. In equation (14) it does not: the price p_t^* varies over time and sits inside both sums, so the welfare value of a unit of emission rights in period t depends on what the carbon price happens to be in that period. A generous allocation in a period of low prices is worth less than the same allocation in a period of high prices. The upshot is that there is no direct correspondence between a country's intertemporal carbon budget R_i and its welfare. The timing of the allocation matters, and the timing depends on the full price trajectory $(p_t^*)_t$.

Special case 1: homothetic prices and fixed budget shares. Suppose the autarky price in each country is always proportional to the global benchmark price: $p_{it} = \pi_i p_t^*$ for a country-specific scalar π_i . Suppose also that country i receives a fixed share of the global emission budget in every period: $r_{it} = \rho_i r_t$, where r_t is the aggregate global allocation. Under these two assumptions, equation (14) simplifies to:

$$\rho_i \mathcal{B} - \sum_t e_{it}^* p_t^* \beta_t \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* p_t^* \beta_t, \quad (15)$$

where $\mathcal{B} = \sum_t r_t \beta_t p_t^*$ is a price-weighted sum of the global allocation. The formula is analogous to the static case but contains $\mathcal{B}$, which requires knowledge of the full price trajectory $(p_t^*)_t$. The equivalence holds, but it is not analytically tractable without a model.

Special case 2: the Hotelling case. In a world with a fixed aggregate carbon budget, perfect foresight, and efficient capital markets, the carbon price rises at the rate of return on capital. When we discount at that same rate, the discounted price $\beta_t p_t^*$ is constant at p_0 in every period. This means p_t^* factors out of the discounted sums in equation (14), and the equivalence reduces to:

$$\boxed{R_i - E_i^* \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^*}, \quad (16)$$

where $E_i^* = \sum_t e_{it}^*$ is country i 's total discounted emissions under the benchmark price trajectory. This is structurally identical to the static formula (9): the intertemporal carbon budget R_i plays the role of the single-period rights allocation, and the homothetic autarky price factor π_i plays the role of p_i/p^* . The timing of the rights allocation no longer matters, because every period carries the same discounted price.

The Hotelling case is the best possible setting for the equivalence: it requires the most favourable conditions (perfect foresight, efficient capital markets, a fixed global budget,

and homothetic autarky prices). If the approximation fails even here, it will fail everywhere. This is why Section 4 uses equation (16) as the empirical benchmark: the simulations ask not whether the Hotelling formula holds in theory, but how far the numerically computed welfare effects deviate from what the formula predicts, and what those deviations reveal about the underlying mechanisms.

4 Numerical comparison between autarky and uniform price with transfers

The static equivalence result and the Hotelling dynamic extension provide clean analytical predictions, but both rest on approximations and structural assumptions (constant elasticities, homothetic price trajectories, and perfect capital markets) that a calibrated model doesn't need to satisfy. Whether the formula $r_i \approx (1 + (1 - p_i/p^*)\varepsilon^*)e_i^*$ accurately predicts the non-losing rights allocation in a full dynamic general equilibrium or not is an empirical question. To answer it, we require a model that (i) captures the non-linear MAC curves that make the Taylor approximation potentially inaccurate, (ii) allows autarky prices and global prices to evolve along distinct trajectories over a multi-decade horizon, and (iii) disaggregates welfare at the sub-national level so that distributional claims can be evaluated precisely. A representative-agent IAM satisfies (i) and (ii) but not (iii); we therefore use the Nested Inequalities Climate Economy (NICE) model (Young-Brun et al. 2025), which extends the DICE framework to disaggregate regional populations into income quintiles. The model evaluated a climate policy timeline spanning from 2030 to 2100. To align with the Paris Agreement, the baseline scenario used a globally calibrated carbon price pathway (p_t^*) designed to enforce a strict 1.8°C global carbon budget, which requires reaching net-zero global emissions by 2080 and achieving net-negative emissions thereafter. To assess the macroeconomic impacts of different policy regimes, the model calculated the Net Present Value (NPV) of both GDP over time and Equally-Distributed Equivalent (EDE) consumption, using a discount rate of 3% and an Atkinson inequality parameter of 1.5.

4.1 Modelling the scenarios

4.1.1 Scenario 1: Autarky baseline

The first scenario simulated an autarky regime where climate policy is entirely decentralized. In this baseline, the target country i sets an independent domestic carbon price proportional to the global target price, defined by the price factor π_i such that $p_i = \pi_i \times p^*$. To enforce the mathematical requirement of true autarky, all international revenue pooling and transfers were strictly disabled within the model, ensuring that revenues from the domestic carbon tax were recycled entirely within the country's own borders. A Negishi welfare-weighting recycling scheme was applied, meaning that financial

transfers are inversely proportional to the marginal utility of consumption. Mathematically, because marginal utility diminishes with income, this scheme allocates a larger absolute share of carbon tax revenues to the wealthiest quintiles and a smaller share to the poorest. This deliberate configuration neutralizes the mechanical reduction in domestic inequality that standard lump-sum recycling would produce, allowing the analysis to isolate the pure macroeconomic efficiency effects of the climate policy on the country's aggregate GDP and consumption. Furthermore, to maintain the global emissions cap across the simulation, the model dynamically calculated the required carbon price for the Rest of the World (RoW), denoted as p_{-i} , using the global price identity:

$$p^* = \omega_i p_i + (1 - \omega_i) p_{-i} \quad (17)$$

where ω_i is country i 's dynamic share of global baseline emissions ($\omega_i = e_i/E^*$). From this, we derive the analytical approximation $p_{-i} = p^* \times (E^* - e_i \pi_i)/e_{-i}$.

However, this identity assumes a linear relationship between carbon prices and emissions, implying that averaging prices based on emission weights results in the exact same global abatement. In the NICE model, the marginal abatement cost (MAC) curve is highly non-linear, and the emission shares are endogenous to the tax itself. Hence, running the model with this naive linear approximation would result in the RoW reducing its emissions by an incorrect magnitude, causing the sum of global emissions ($E_i + E_{-i}$) to breach the climate cap.

To strictly maintain the global carbon budget across the simulation, an iterative root-finding algorithm was implemented to solve for the true p_{-i} . The model dynamically updated the RoW price based on the residual gap between simulated global emissions and the target cap. A Newton-Raphson method was utilized for rapid convergence, with a bisection algorithm serving as a fallback for extreme boundary cases where Newton iterations stalled. The algorithm iterated until the relative error between global emissions and the cap fell strictly below 0.1%. This threshold ensures mathematical precision, as in absolute terms, 0.1% represents a negligible fraction of a gigaton of CO₂, securing the integrity of the carbon budget without risking infinite computational loops.

4.1.2 Scenario 2: Uniform price with varying rights

The second scenario simulated a uniform global cap-and-trade system where all regions face the global carbon price p_t^* , but the allocation of emission rights varies. The target country i was granted a share s_i of the total global carbon budget E^* proportional to its share of the global population (pop_i), scaled by a rights multiplier ρ_i . This multiplier represents the ratio of country i 's share of global emission rights over its share of the world population (i.e., the distance to a pure per-capita rights allocation).

The core analytical objective of this exercise is to systematically vary ρ_i for the target country to observe the corresponding macroeconomic welfare shifts. Thus, their rights must strictly follow this population-based formula by construction. However, to prevent the macroeconomic collapse of heavy emitters in the aggregate RoW group and maintain

global mathematical consistency, the model deliberately applied an asymmetric allocation rule. While the target country's rights scale with the multiplier, the remaining global carbon budget was distributed to the RoW proportionally based on their baseline emissions (grandfathering).

Indeed, if population-based rights were symmetrically applied to the RoW, the sum of all regional rights would violate the global cap too often (summing to either more or less than E^*), rendering the scenarios incoherent. Grandfathering distributes the residual cap to exactly match E^* , while providing a status-quo-neutral baseline allocation for the RoW regions, which are not the primary variable of interest in the analysis.

4.1.3 Methodological extensions

To directly compare the two regimes, an automated grid search was executed over the (ρ_i, π_i) parameter space for a subset of eight representative entities: the United States, Russia, the European Union (27 states), India, China, Turkey, the Democratic Republic of Congo, and Nigeria. For each combination, we then computed the relative variation in the NPV of GDP and EDE consumption:

$$((NPV_{\text{uniform}} - NPV_{\text{autarky}}) / NPV_{\text{autarky}}) \times 100 \quad (18)$$

To execute this grid search across highly varied parameters, a critical extension to the standard NICE framework was implemented. We allowed for abatement rates exceeding 100% ($\mu_i > 1$). Standard IAMs generally constrain $\mu_i \in [0, 1]$, but enforcing this limit causes computational crashes under extreme policy parameters. Allowing $\mu_i > 1$ mathematically represents the deployment of Carbon Dioxide Removal (CDR) technologies, implying the region is extracting more CO₂ than it emits. To model these costs, the standard nonlinear MAC function was extrapolated beyond the 100% boundary.

Finally, we introduced a viability filter to flag economic and mathematical policy extremes. Two boundary conditions were evaluated continuously across every year of the 2030-2100 timeline:

1. **Market-distorting subsidies** ($\omega_i \times \pi_i > 1$): This condition flags scenarios where the target country claims more than 100% of the global price burden. To compensate for a heavily emitting nation applying a massive π_i multiplier, the model is mathematically forced to push the RoW price below zero ($p_{-i} < 0$). Economically, this acts as a fossil-fuel subsidy for the RoW, incentivizing emissions.
2. **Negative emission rights** ($\rho_i > 1 / pop_i$): This condition flags scenarios where granting the target country its rights multiplier uses up more than 100% of the total global carbon budget. This forces the RoW into negative emission rights. This boundary is particularly relevant for nations with large populations. For example, countries holding more than 10% of the global population (like China or India) inherently trigger this flag when exploring multipliers $\rho_i > 10$.

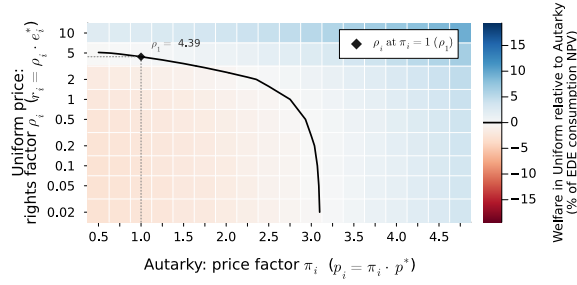
4.2 Results

To visualise welfare differences between the two scenarios, we construct one heatmap per country mapping the relative NPV of consumption EDE onto the two-dimensional parameter space (π_i, ρ_i) . The x-axis gives the autarky price factor π_i , the y-axis (on a log scale) gives the rights multiplier ρ_i , and the colour encodes the welfare gain of the uniform regime relative to autarky, expressed as a percentage of the NPV of EDE consumption. Blue cells indicate that the uniform price with transfers dominates autarky, red cells indicate the reverse. The empirical indifference curve separates the two regions and is the central object of interest. It shows, for any autarky price π_i , the minimum rights allocation ρ_i that makes country i weakly prefer the uniform regime, and its intercept at $\pi_i = 1$ gives the non-losing rights multiplier ρ_1 .

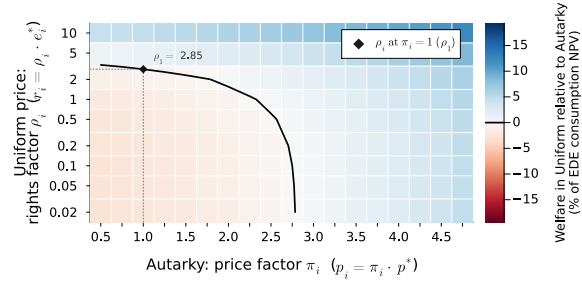
Table 1 reports ρ_1 alongside each country's emission share, population share, per-capita emissions, and a predicted ρ_1 derived from the static formula. Evaluating equation (10) at $\pi_i = 1 \iff p_i = p^*$ gives $r_i = e_i^*$, so the predicted non-losing allocation equals the country's equilibrium emissions under the uniform price expressed as a multiple of its population share of the global budget. Under the Hotelling approximation this simplifies to the ratio of emission share to population share, which is algebraically equivalent to per-capita emissions relative to the world average, hence the near-identical values in the last two columns of Table 1 for most countries. The central empirical finding is that the static formula predicts the simulated non-losing allocation well for the large majority of countries: the deviation between predicted and actual ρ_1 is below 17% in absolute terms for the USA (-1.1%), Russia (-1.4%), India (+4.6%), Turkey (+10.9%), and Nigeria (-16.7%), and while the DRC's percentage deviation is larger (-33%), the underlying ρ_1 values are so small (0.03 predicted versus 0.02 actual) that the absolute gap is negligible. Only two countries show a deviation large enough to warrant separate explanation: China (-29.5%) and the EU27 (-23.3%). The discussion below first establishes why the static formula performs well across most of the sample, then turns to the dynamic general-equilibrium mechanisms behind the more moderate departures observed for China and the EU27.

4.3 Countries where the static formula holds

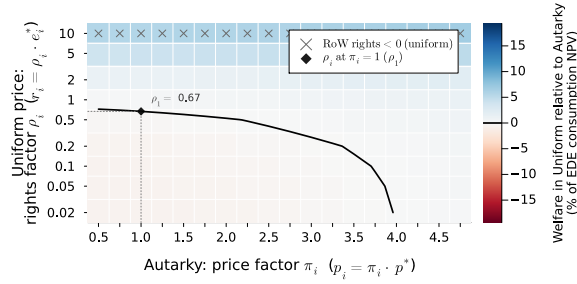
For six of the eight countries in the sample (USA, Russia, India, Turkey, Nigeria, and the DRC) the actual ρ_1 is within a small margin of the static prediction. The non-losing allocation scales with per-capita emissions relative to the world average holds as a first approximation across a rather wide range of income levels and emission intensities. The USA (-1.1%) and Russia (-1.4%) match almost exactly, consistent with both being high-income, high-per-capita-emissions countries. India (+4.6%) and Turkey (+10.9%) deviate a little more, but the sign and magnitude of these small gaps are consistent with both countries' baseline emissions still growing over the simulation horizon: equilibrium emissions under the uniform price run slightly above the 2030 snapshot used in the static prediction, but not by enough to move either country out of the predictive power of the



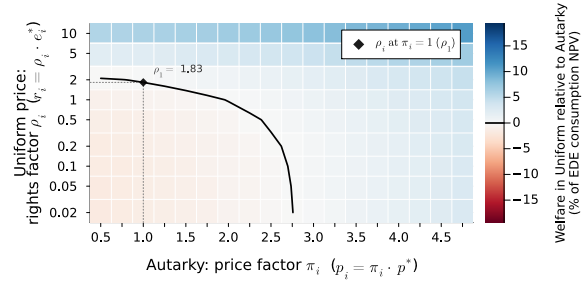
(a) United States (USA)



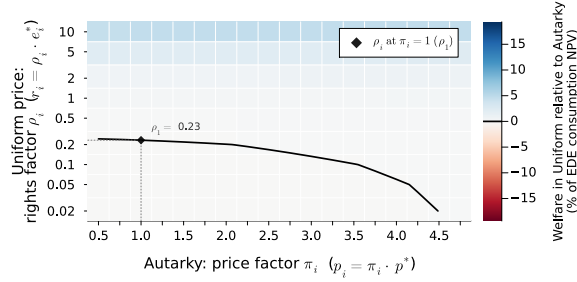
(b) Russia (RUS)



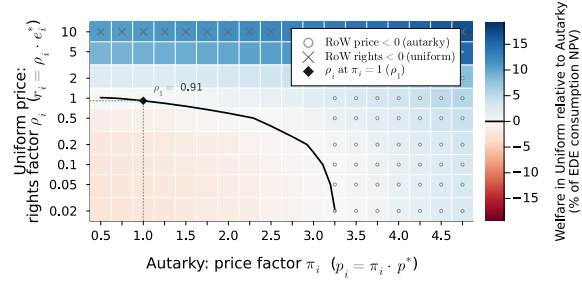
(c) China (CHN)



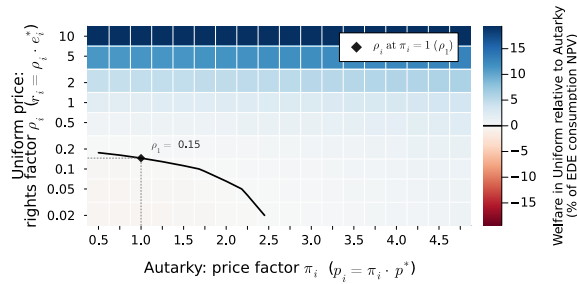
(d) Turkey (TUR)



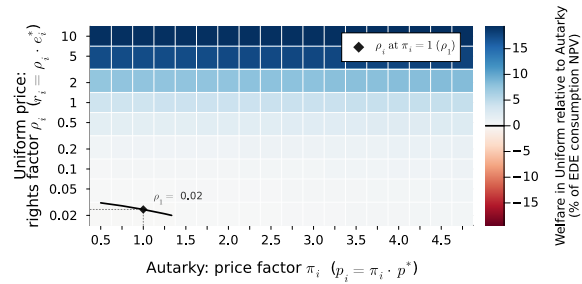
(e) European Union (EU27)



(f) India (IND)



(g) Nigeria (NGA)



(h) Democratic Republic of Congo (COD)

Figure 1: Welfare gain (consumption EDE) from Uniform Price scenario to Autarky scenario

Table 1: Non-losing rights allocation factor ρ_i at $\pi_i = 1$ (ρ_1) by country

Country	Emission share in 2030 (%)	Population share in 2030 (%)	Emissions p.c in 2030 (tCO ₂)	Predicted ρ_1^*	ρ_1
USA	15.9	4.1	14.12	4.44	4.39
RUS	3.8	1.7	8.26	2.89	2.85
CHN	29.9	17.3	6.34	0.95	0.67
TUR	1.2	1.1	4.18	1.65	1.83
EU27	5.6	5.2	3.97	0.3	0.23
IND	8.3	17.8	1.72	0.87	0.91
NGA	0.4	3.1	0.46	0.18	0.15
COD	0.0	1.4	0.05	0.03	0.02

* The predicted ρ_1 equals the country's emission share divided by its population share, i.e., $(e_i/E^*)/(pop_i/Pop)$, computed as an NPV, in the context of the autarky scenario with $\pi_i = 1$.

formula.

Nigeria (-16.7%) is the largest percentage deviation among this group, but the underlying values are small enough that the absolute gap remains modest.

Among the conforming countries, Nigeria and the DRC illustrate the Diamond–Mirrlees efficiency argument the best. Both have negligible per-capita emissions and very limited domestic abatement capacity, so the autarky price is almost irrelevant to their welfare. What matters is the rights allocation: even a small positive ρ_i generates a transfer that is large relative to national income, because the country holds rights it can sell at the full global price p^* without needing to abate anything. For the DRC at $\rho_i = 0.5$, Table 3 shows transfer revenues of around \$9 billion in 2030, approximately 13% of its current GDP, against actual emissions of just 0.005 GtCO₂. No autarky price comes close to delivering this outcome. For the poorest countries, differentiated prices offer very small advantages, as the transfer from trading dominates in most cases.

Table 2: Autarky scenario for the United States ($\pi = 2$)

Variable	2030	2040	2050	2060	2070	2080	2090	2100
p^* (USD/tCO ₂)	42.6	112.9	205.9	285.7	401.6	480.9	476.2	471.4
p_i (USD/tCO ₂)	85.2	225.8	411.7	571.4	803.3	961.9	952.3	942.8
E_i (GtCO ₂)	4.31	2.58	0.63	-0.53	-1.75	-2.22	-1.86	-1.52

Table 3: Uniform price scenario for the Democratic Republic of Congo ($\rho = 0.5$)

Variable	2030	2040	2050	2060	2070	2080	2090	2100
p^* (USD/tCO ₂)	42.6	112.9	205.9	285.7	401.6	480.9	476.2	471.4
r_i (GtCO ₂)	0.221	0.185	0.131	0.099	0.043	0.0	0.0	0.0
e_i (GtCO ₂)	0.005	0.007	0.008	0.008	0.004	0.0	0.0	0.0
Transfer (bn USD)	9.2	20.1	25.4	25.9	15.4	0.0	0.0	0.0

4.4 China and the EU27: moderate deviation from the prediction

China ($\rho_1 = 0.67$, predicted 0.95) and the EU27 ($\rho_1 = 0.23$, predicted 0.3) both need somewhat fewer rights than the static formula suggests to be indifferent between the two regimes. They show deviations of -29.5% and -23.3% respectively. These are the two largest gaps in the sample, but they are still quite moderate. Both countries benefit from joining the uniform-price regime with a rights allocation reasonably close to the per-capita-based prediction, for two different reasons.

For China, the deviation shows that, in this context, free-riding is not necessarily beneficial at such a scale. With nearly 30% of global emissions in 2030, a low Chinese domestic price forces a sizeable compensating adjustment on the rest of the world to maintain the global cap, which feeds back negatively through trade and capital markets and offsets part of the domestic benefit of setting $\pi_i < 1$. The heatmap in Figure 1 shows the uniform-price regime dominating autarky over a wide part of the tested parameter space for China, and China is willing to accept a somewhat below-proportional rights allocation to access the global market.

For the European Union, the mechanism is different. Indeed, it has already implemented a lot of its cheap abatement through the ETS and domestic climate regulation. In autarky, meeting the 1.8°C budget forces it onto the expensive part of that curve at every price level, making domestic-only decarbonisation costly. Access to the international market, where low-cost reductions remain available in developing countries, is therefore valuable even at a somewhat reduced rights allocation. The EU’s deviation from the static prediction is a modest illustration of the Diamond–Mirrlees argument among high-income countries: when domestic abatement options are largely exhausted, the efficiency gain from a uniform price with international trading can compensate for a below-proportional initial allocation, though here the gap remains contained rather than extreme.

Autarky is costly relative to the uniform-price regime across nearly the full range of domestic price factors π_i tested, rather than only at extreme values. Access to the international market, where lower-cost reductions remain available in other regions, is therefore valuable to the EU27 even at a below-proportional rights allocation. The EU’s deviation from the static prediction is a modest illustration of the Diamond–Mirrlees argument

among high-income countries: the efficiency gain from a uniform price with international trading can compensate for a below-proportional initial allocation, though here the gap remains contained rather than extreme.

5 Equivalent rights for recent price proposals

Two recent proposals offer differentiated carbon-price schedules that can be fed directly into the NICE model as country-level π_i inputs. [Wolfram et al. \(2025\)](#) propose a voluntary coalition of countries that would align on a tiered carbon-pricing schedule, with higher-income members facing higher price floors, paired with a shared carbon border adjustment mechanism. The report estimates that the coalition could raise close to \$200 billion per year in domestic carbon-tax revenue. [Banerjee et al. \(2025\)](#) propose a “grand bargain” in which low-and middle-income countries become eligible for climate-damage compensation, conditional on adopting a graduated domestic carbon price: \$10/tCO₂ for low-income countries, \$30 for lower-middle-income countries, and \$50 for upper-middle-income countries. Because the two proposals define π_i differently by construction (a shared coalition schedule in one case, an income-conditional compensation mechanism in the other), the resulting country-level price factors, and hence the ρ needed to match them, diverge substantially across proposals for the same country.

Applying the two proposals’ price schedules as π_i inputs to the NICE model yields non-losing rights allocation each country would need to be indifferent to trading its proposal-specific price for uniform pricing, reported in Table 4.

6 Concluding remarks

Across all eight countries, the uniform-price regime weakly dominates autarky on efficiency grounds, consistent with the Diamond–Mirrlees theorem. The required non-losing rights allocation varies enormously, from $\rho_1 \approx 0.02$ for the DRC to $\rho_1 \approx 4.39$ for the USA, a range of more than two orders of magnitude. The static formula explains this variation well for most countries, with larger deviations for China and the EU27

The political challenge is therefore not about efficiency but about agreeing on the rights allocation. However, the red region below the indifference curve is barely visible in most heatmaps, rarely exceeding a few percentage points of EDE consumption NPV, while the blue region above it reaches the scale ceiling for countries like Nigeria and the DRC. This asymmetry between potential gains and losses is a visual confirmation of the Diamond–Mirrlees argument: the efficiency cost of being slightly below the indifference curve is small, while the efficiency gain from a generous rights allocation can be large, particularly for low-emitting countries with negligible domestic abatement capacity. It also means that there is room for negotiated compromise without incurring large efficiency losses.

Table 4: Uniform-price rights ratio ρ_i equivalent to the Wolfram and Duflo autarky scenarios

Country	Wolfram			Duflo		
	ΔT	Welfare	ρ_{Wolfram}	ΔT	Welfare	ρ_{Duflo}
CHN	+7.4	529.46	1.306	+7.4	530.02	1.514
COD				-32.6	165.56	< 0.0001
EU27	+32.4	1247.15	< 0.0001	-42.6	1247.5	0.043
IND	-17.6	255.88	1.266	-12.6	254.58	0.836
NGA				-12.6	265.86	< 0.0001
RUS				-42.6	857.24	8.043
TUR				+7.4	760.05	0.574
USA				-42.6	1171.3	11.04
World rights reallocation (%)		50.19%			48.25%	
World temp. 2100		2.14 $\rightarrow$ 1.757			1.984 $\rightarrow$ 1.757	
Autarky $\rightarrow$ uniform ($^{\circ}\text{C}$)						
World welfare gain (%)		0.26%			3.2%	

Note: ' $<$ ' (resp. ' $>$ ') indicates that welfare under uniform pricing already exceeds (resp. never reaches) the autarky benchmark over the tested raw ρ range; the true equivalent ratio lies below (resp. above) the reported bound. ΔT represents the difference between the domestic autarky tax and the global uniform price in 2030.

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